Digital Forensics and Incident Response (DFIR) Report: Endpoint Live Response Analysis

Incident Reference: IR-202610-002

Analyzed Asset: `dataorg-dv-28m` (Azure Virtual Machine)

Report Type: Live Response Artifact Comparison

1. Executive Summary

Comparison of pre-incident and post-incident Microsoft Defender for Endpoint (MDE) live response packages collected from `dataorg-dv-28m` confirms a successful interactive host compromise. The post-breach environment exhibits definitive indicators of post-exploitation activity, specifically the installation of a masqueraded persistence mechanism and the execution of native Windows binaries indicative of defense evasion. This activity correlates with the previously established Remote Desktop Protocol (RDP) brute-force timeline.

2. Baseline Determination & Methodology

- **Baseline Package:** `Pre-breach Investigation Collection.zip` (259 artifacts)
- **Comparison Package:** `Post-Breach Investigation Collection.zip` (381 artifacts)
- **Justification:** The `Pre-breach` package is utilized as the known-good baseline based on collection nomenclature and a lower artifact volume, representing the system state prior to the introduction of adversary tooling and post-exploitation execution.

3. Probable Incident Type

Interactive Post-Exploitation / Ransomware Precursor. The evidence does not indicate a fully automated worm or opportunistic script at the OS level. The specific sequence of deploying a disguised service (`defenderupdatesvc`) combined with the manual execution of command-line utilities to manipulate audit policies (`auditpol.exe`) and terminate processes (`taskkill.exe`) is characteristic of human-operated ransomware operators or initial access brokers (IABs) preparing the environment for a final payload.

4. Artifact Delta Analysis

Artifact Category	Identified Delta	Change Type	Assessment	Evidence Source
Installed Services	`defenderupdatesvc`	ADDED	**Suspicious**	`Services/Services.csv`
Prefetch (Execution)	`AUDITPOL.EXE-5C071DAC.pf`	ADDED	**Suspicious**	`Prefetch/AUDITPOL.EXE-*.pf`
Prefetch (Execution)	`TASKKILL.EXE-ECD4FD3D.pf`	ADDED	**Suspicious**	`Prefetch/TASKKILL.EXE-*.pf`
Network Connections	Ext. IPs (e.g., `185.44.66.153`, `23.222.201.247`)	ADDED	**Suspicious**	`ActiveNetConnections.txt`
Prefetch (Execution)	`ONEDRIVE.SYNC.SERVICE.EXE`	ADDED	Benign	`Prefetch/ONEDRIVE*.pf`
Running Processes	`mousocoreworker.exe`, `gc_worker.exe`	ADDED	Benign	`Processes/Processes.csv`
Scheduled Tasks	`\OneDrive...`, `\Softlanding...`	ADDED	Benign	`ScheduledTasks.csv`
Local Users/Groups	No new privileged accounts created	UNCHANGED	Benign	`LocalGroups.txt`, `QueryUser.txt`

5. Prioritized Indicators of Compromise (IoCs) & ATT&CK Mapping

1. **T1543.003 (Windows Service) / T1036 (Masquerading):** The creation of the `defenderupdatesvc` service. The adversary deliberately chose a naming convention mirroring legitimate Microsoft Defender components to bypass casual analyst scrutiny and ensure persistence across reboots.
2. **T1562.002 (Disable Windows Event Logging):** The presence of `AUDITPOL.EXE` in the Prefetch cache. This indicates the adversary interacted with the Windows Audit Policy command-line tool, highly likely attempting to clear event logs (`auditpol /clear`) or disable granular success/failure logging to cover their tracks.

3. **T1562.001 (Disable or Modify Tools):** The execution of `TASKKILL.EXE`. The threat actor utilized this utility to forcefully terminate running processes, typically targeting endpoint detection and response (EDR) sensors, antivirus agents, or database services locking critical files.

6. Gaps and Limitations

- **Lack of Command-Line Telemetry:** Prefetch files confirm the execution of `auditpol.exe` and `taskkill.exe`, but MDE live response packages do not capture the exact arguments used. We cannot definitively state which processes were killed or which specific audit policies were disabled without querying the `DeviceProcessEvents` table in Advanced Hunting.
- **Missing Binary Context:** The `Services.csv` artifact identifies the rogue service name but lacks the file path and cryptographic hashes of the underlying executable.
- **Temporal Resolution:** Live response packages represent a static snapshot of the system state at the time of collection. Establishing the exact chronological sequence of post-exploitation commands requires a full forensic timeline (parsing the `$MFT`, Windows Event Logs, and USN Journal), which is outside the scope of a standard MDE package.

7. Recommended Immediate Actions

1. **Asset Containment:** If not already completed, enforce strict network isolation on `dataorg-dv-28m` via Azure Network Security Groups (NSGs) to sever active Command and Control (C2) and prevent lateral movement.
2. **Payload Acquisition:** Query the registry (`HKLM\SYSTEM\CurrentControlSet\Services\defenderupdatesvc`) or use MDE Live Response to extract the `ImagePath` for the malicious service. Pull this binary for reverse engineering and generate file hashes for enterprise-wide sweeps.
3. **Account Remediation:** Treat the local `administrator` account as fully compromised. Initiate a forced password reset and enforce MFA for any administrative access to the Azure tenant and remaining virtual machines.
4. **Perimeter Blocking:** Ingest the anomalous IP addresses identified in the active network connections (e.g., `185.44.66.153`) into edge firewall blocklists.